Reading the Reader: A Modular Platform for Auditable Gaze-Contingent Reading Experiments

Sachin Baral
Technical University of Denmark
Kongens Lyngby, Denmark

Satish Gurung
Technical University of Denmark
Kongens Lyngby, Denmark

Abstract

Gaze-contingent reading experiments connect noisy eye-tracking signals to changing text, while requiring researcher control and a record of how changes occurred. We present Reading the Reader, a modular platform separating sensing, eye-movement analysis, decision proposals, and typographic interventions. A researcher console supports manual changes and supervised proposals; a participant reader maps gaze to rendered text and reports context-restoration outcomes. We evaluate the implementation through automated tests, eight recorded sessions with four participants, a separate advisory-provider run, and browser geometry records. The sessions contain 175,860 gaze samples at a mean session rate of 89.89 Hz; participant-client diagnostic round-trip time has a 3 ms 95th percentile. These measurements establish capture and transport behaviour in a single-host deployment, not complete gaze-to-display latency. Auditing also identifies repeated derived-event payloads and restoration failures. The contribution is an inspectable experimental workflow and a technical account of its capabilities and measurement limits.

CCS Concepts

• **Human-centered computing → User interface toolkits**.

Keywords

eye tracking, gaze-contingent reading, adaptive typography, research software, session provenance

1 Introduction

A researcher testing adaptive typography must connect several operations that are usually considered separately. An eye tracker supplies coordinates; an analysis procedure interprets eye movements; a decision rule proposes a change; and a reader application changes the text. Increasing font size can then move the very word used to trigger the change. The researcher needs to know which text position was estimated, which proposal was accepted, when the presentation state changed, and whether the intended reading position remained available. A functioning display update alone does not answer these questions.

Recent reading interfaces show why this workflow matters. Context-preservation designs help readers relocate text after typographic changes [Jensen et al. 2025], and gaze-based word highlighting has been evaluated as reading support [Rummens and Beier 2026]. Such interventions depend on choices about mapping, timing, and presentation. Exchanging a detector or trying a different intervention policy should not require rebuilding the entire experiment or losing the relationship between its inputs and outcomes.

General experiment builders provide foundations for stimulus presentation, hardware integration, and custom procedures [Mathôt et al. 2012; Peirce et al. 2019]. Our focus is the concrete workflow of an operator running a gaze-contingent reading experiment: configure an analysis provider, observe its output, apply or approve a typographic change, and inspect the resulting session. We bring these activities together in a reading-specific instrument whose extension boundaries are connected to its experimental record.

We present *Reading the Reader*, a platform with a participant reader, a researcher console, and a backend that owns session state. Its four principal contracts separate sensing, eye-movement analysis, decision proposals, and interventions. This separation permits a manual experiment to use external analysis without delegating presentation control to that analysis. An advisory decision provider can subsequently be introduced while leaving the researcher responsible for approval. The same session record retains gaze, derived events, proposals, applied interventions, and browser-reported restoration outcomes.

We investigate how this design supports configurable experiments and what its existing technical evidence establishes. The evaluation combines source inspection and automated tests with eight Tobii-backed sessions, a separate advisory run, and a browser displacement sweep. It also examines the quality of the records used to evaluate the platform. This audit is necessary because repeated derived events or mismatched anchors can change the interpretation of otherwise plausible results.

The paper makes three contributions:

(1) A working architecture and operator workflow that separates analysis output, intervention proposals, and committed presentation changes within gaze-contingent reading experiments.
(2) An integration account covering reference providers, a connector to collaborator-developed reading analysis, and context restoration with explicit outcome reporting.
(3) A technical evaluation with defined measurement endpoints, event-integrity checks, and documented limits on behavioural and geometric inference.

The recorded sessions demonstrate the instrument in use. They do not establish improved reading performance, the effectiveness of a learned decision policy, or complete gaze-to-display latency. Keeping those questions distinct makes the platform's present contribution assessable and identifies the measurements needed for subsequent intervention studies.

2 Related Work

2.1 Gaze-contingent reading and context preservation

Jensen et al. [2025] studied context preservation in an adaptive reader using Popup, Undo, Notification, and Gradual interventions, with word-, sentence-, and paragraph-level highlighting. Their 22-participant within-subjects study motivates treating the reader's recovery after a layout change as part of intervention design. It provides the closest application context for this work. We examine the instrument through which researchers configure, control, and record an experimental loop around such interventions.

Rummens and Beier [2026] evaluated gaze-based word colouring with 70 Danish second graders reading aloud. Their reported gains in reading efficiency coexisted with a preference for static text. This distinction matters for adaptive-reading research: a gaze-derived measure, task performance, and subjective preference describe different outcomes. It motivates separating the technical behaviour of the intervention system from the reader's response. Our evaluation concerns that technical behaviour; the exploratory movement proxy is defined separately from comprehension and preference.

Medan and Pelman [2024] explored a different interaction in *New Line*: gaze controls scrolling through a continuous line and emphasises the attended word. Their pilot illustrates that gaze-contingent reading can alter navigation as well as typography. Ilyas et al. [2025] compared reading of human- and AI-generated texts using eye-tracking measures. These applications motivate retaining both stimulus identity and the processing history behind derived measures. They address different experimental questions from the modular orchestration studied here.

2.2 Assigning gaze to changing text

A live reader needs a mapping from noisy coordinates to the words currently rendered. Line transitions and regressions complicate that mapping, especially when the layout changes. Kaltenberger et al. [2026] present CONF-LA, a confidence-based approach to real-time line assignment that can defer uncertain assignments. Their work highlights the distinction between assigning a line and representing uncertainty about that assignment. Our browser uses a geometric heuristic with a preference for the current line, rather than CONF-LA or a calibrated confidence model. Replacing or validating this mapper remains a separate algorithmic task.

2.3 Experiment platforms and integration scope

PsychoPy supports both graphical experiment construction and scripting, with facilities for hardware input and controlled stimulus presentation [Peirce et al. 2019]. Its ioHub documentation describes a common eye-tracker interface across multiple devices and includes simulated gaze [PsychoPy contributors 2026]. OpenSesame similarly combines a graphical experiment builder, Python scripting, and plug-ins, including eye-tracking support [Mathôt et al. 2012]. These systems establish that replaceable hardware and programmable experiments are existing capabilities.

Our design concentrates on an integrated reading workflow: the researcher can inspect an external provider's proposal separately from the presentation change, and the browser can report what happened to a reading anchor after that change. General-purpose experiment builders supply infrastructure for programming custom procedures. The present implementation supplies these reading-specific operations together, including their shared session state and export. Its evaluation therefore examines the implemented workflow and how well the resulting records support inspection of a completed experiment.

3 Platform and Design Rationale

3.1 Two users and one authoritative session

The participant reads paginated text in a browser. A separate researcher console configures the session, monitors gaze and analysis, applies typography changes, and resolves advisory proposals. A .NET backend owns the session state and distributes updates to both surfaces. Figure 1 shows the functional boundaries. The browsers share session state through the backend; the participant surface does not take commands directly from a model process.

A typical experiment begins with content selection and a configured sensing source. The researcher performs calibration and validation, starts recording, and observes the analysis alongside the current text. In manual operation, the researcher selects a presentation change from the console. In advisory operation, a decision provider supplies a proposal that the researcher can resolve. The backend validates the requested intervention and determines when to commit it. The participant reader responds to the resulting state update and, if preservation is enabled, attempts to restore the reading context. The researcher can then export the record and inspect the event sequence in replay.

This ownership model separates observations from authority. An analysis provider may report eye-movement features without being allowed to mutate presentation state. A proposal can exist without being applied. A committed change can exist even if the browser later reports a degraded restoration. Keeping these states distinct permits the researcher to locate a failure at the relevant stage.

3.2 Four contracts and their extension boundaries

Sensing. The sensing adapter translates device callbacks into domain gaze samples with timestamps and validity information. The implementation includes a Tobii path and mouse-based input for development. The current Tobii SDK binding is implemented for Windows; the recorded hardware evidence covers one Tobii device model. The acquisition path updates the latest sample without taking the session lifecycle semaphore, while recording uses a separate history lock. This separates acquisition updates from lifecycle coordination while serialising access to recorded history.

Eye-movement analysis. An analysis coordinator selects a configured strategy. A built-in strategy interprets token observations, while an external strategy accepts provider snapshots. In the participant browser, word boxes are derived from the live document layout. A geometric line score combines vertical distance and line-centre distance, with a fixed preference for the current line; a word is then selected within the candidate line. Layout mutations, resize events, and scrolling refresh the geometry. This makes the mapper responsive to reflow, but it does not establish token-level accuracy under noisy gaze.

Decision proposals. The decision coordinator separates manual control from configured decision strategies. An external provider receives session context and returns proposals through a message

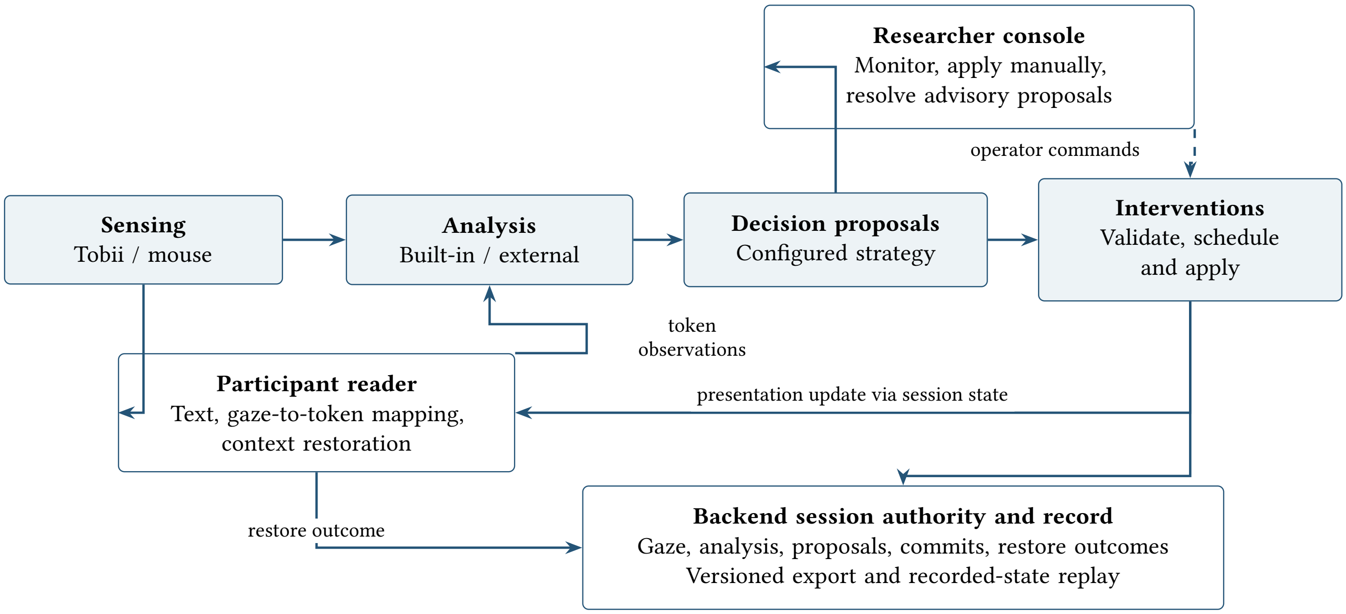

Figure 1: Functional view of the experiment loop. The backend owns session state and mediates provider and browser messages. Solid arrows show the principal observation and presentation flow; the dashed arrow shows operator control. The record includes events from all stages. Boxes denote logical responsibilities, not separate deployment units.

protocol. Advisory mode exposes proposals for researcher resolution; autonomous handling is also implemented and covered by lifecycle tests. Proposal identity and status are retained separately from the intervention event. The reference decision provider is an integration example, not a validated model of reading difficulty.

Interventions. An intervention module supplies a descriptor, validation, and execution behaviour. Eight built-in module kinds cover font family, font size, line width, line height, letter spacing, theme, palette, and edit locking. Layout changes can be committed immediately or queued for a configured reading boundary. Automated step and cooldown constraints apply in the relevant decision path; manual commands are exempt from those automated limits. New modules that use the existing presentation vocabulary can reuse this machinery. A new rendering behaviour may also require frontend changes, so backend registration alone is not a universal extension mechanism.

These are logical extension boundaries. Several strategies and interventions reside in the same application assembly. Project dependencies point inward, and tests guard specified assembly references, but the four contracts are not four independently deployed services or universally enforced compiler boundaries.

3.3 Integrating an external analysis provider

The repository contains a reference analysis provider, a reference decision provider, and a connector to a collaborator-developed reading-analysis pipeline. The connector consumes raw gaze, translates it into the collaborator pipeline's input representation, and sends analysis snapshots back to the platform. It reconciles device timestamps with an epoch-based time reference and associates available token observations with processed events. The current integration also contains transport-related additions for cross-word saccades. The evidence therefore concerns an adapted integration with collaborator software, not an unchanged third-party program or an unaided external developer study.

This example exposes a useful boundary between acquisition and analysis cadence. The connector's default submission threshold is 180 samples, approximately two seconds of input at 90 Hz. That default must not be confused with a measured historical configuration or with the arrival rate of individual gaze samples. A backend can receive a recent gaze sample while the most recent analysis snapshot still summarises an earlier window.

Provider availability is also distinct from strategy selection. In the inspected external path, an unavailable provider produces no fresh result; the coordinator does not automatically select the built-in algorithm in response. The operator can observe provider state, but a continuing session alone does not demonstrate successful algorithm fallback. These limits matter when researchers decide what a recording made during a disconnection can support.

3.4 Restoring reading context after reflow

The browser restoration hook periodically captures a reading anchor. It first uses an active gaze-associated word when available, then falls back to a visible location near 35% of the viewport height, and finally to a coarser block. The current implementation refreshes the anchor every 120 ms and uses a 15 s freshness test. These are implementation parameters, not empirically validated thresholds for reading recovery.

After a presentation change, the hook attempts to locate the anchor in the new layout and reposition the reader. The lookup prefers a sentence wrapper before the individual token and block fallbacks. A temporary context highlight is held for three seconds and fades over the following three seconds. The hook records an outcome after two animation-frame callbacks, including restoration status and geometric diagnostics. Those callbacks schedule browser work; they do not measure the time at which pixels become visible on the display.

Capturing a word and restoring a sentence are related but different operations. A sentence can be successfully relocated while the original word moves within it. Consequently, sentence-anchor residual error and displacement of the original word must be reported separately. The geometry evaluation in Section 5.4 examines existing records with this distinction in mind. The recorded participant sessions used the original restoration behaviour; the revised implementation has only browser-level evidence here.

3.5 Recording, export, and replay

The session record brings together raw gaze, derived fixation and saccade events, proposals, applied interventions, and context-preservation outcomes. Export uses a versioned JSON schema with session metadata and content; timing telemetry is also exported separately. The replay reader reconstructs recorded state at a selected playhead. This lets an investigator inspect the relationship between a proposal, a committed change, and subsequent events without repeating the live session.

Replay reconstructs recorded events rather than rerunning the detector or model. Exact scientific reproduction would additionally require the historical code revision, provider configuration, fonts, browser environment, and processing parameters. The current export is also not a complete configuration timeline: some condition fields are taken from the final session snapshot. We therefore treat event-level provenance and final configuration labels as different sources when interpreting the recordings.

4 Evaluation Methods

We evaluate three questions: which extension and control paths are implemented and exercised; what capture and timing behaviour is supported by the recordings; and what the exported records reveal about intervention outcomes. Evidence comes from four sources: implementation inspection and backend tests, eight reading-session exports with their telemetry, a separate advisory-provider recording, and two browser geometry result files. The latter three are reanalyses of existing thesis artifacts [Baral and Gurung 2026], not newly collected experiments. We keep the sources separate because they exercise different paths and versions.

4.1 Implementation and integration checks

Source inspection follows a sample from sensing through analysis, decisions, intervention execution, browser restoration, and export. We examine the reference providers and collaborator connector to identify what the extension contracts actually require. The backend persistence and runtime test project was rerun during the manuscript review: 101 tests passed, with no failures or skips. Its coverage includes specified assembly dependencies, proposal lifecycles, intervention execution, export serialization, and recovery storage. These checks concern the current checkout; they do not certify the historical device session or browser behaviour.

4.2 Recorded sessions and materials

Four participants each completed two sessions, one with preservation enabled (ON) and one without it (OFF). The thesis describes a convenience sample involving the authors and acquaintances. These are instrument-development sessions, with no population-level efficacy claim. The available records identify two expository articles, *Metal Detectors* and *Nuclear Science*. Each appears twice in each condition, but every participant reads a different article across their two conditions. P1's recorded order is ON then OFF; the other three participants have the reverse order. Random allocation and equivalence of text difficulty are not established by these records.

The reported setup used a Tobii `IS4_Large_Peripheral` tracker and a 2560 × 1440 participant display. Calibration used nine points, followed by validation, and all eight exports record a passing result. The backend and both browser surfaces ran on one host, so browser–backend communication used local loopback. The thesis identifies an external analysis provider for the sessions. All 45 recorded interventions have a manual source and an immediate application boundary; they therefore do not test the benefit of autonomous decisions or boundary-based scheduling.

Intervention counts, timing, and types vary across sessions (Table 1). They include typography and palette changes. Preservation combines positional restoration with a temporary highlight, so a condition comparison cannot isolate highlighting. All sessions contain quiz answers, but the current analysis neither validates a comprehension score nor interprets complete-session duration as reading speed. One OFF session has final provider metadata indicating a rule-based advisory configuration despite recording only manual interventions and no proposals. Because the export takes this metadata from a final snapshot, event source is the basis for the statement about manual changes; configuration history remains unresolved.

Participant-procedure documentation. The thesis's engineering ethics account and its recorded participant evaluation do not supply a consistent consent and institutional-review description. Recruitment details, consent, and the applicable review or exemption determination must be confirmed by the authors before submission. This draft does not infer an approval or exemption from the small sample or local deployment.

4.3 Audit and metric definitions

The manuscript audit selects the two explicit preservation folders, verifies session schema version 7, and joins each telemetry export to its session by identifier and start/end timestamps. Additional root-level advisory exports are analysed separately. A script using the Python standard library produces CSV summaries and a JSON audit with SHA-256 hashes for its 20 inputs. Original recordings are left intact. This inclusion rule avoids a recursive loader treating advisory or telemetry files as additional participants.

For session s, the effective gaze rate is

$$f_s = \frac{N_s - 1}{(t_{s,\text{last}} - t_{s,\text{first}})/10^6}, \tag{1}$$

where t is the device timestamp in microseconds and N_s is the number of timestamped samples. We report the unweighted mean and sample standard deviation across sessions. Gaze validity is the

Table 1: Recorded sessions. M = *Metal Detectors*; N = *Nuclear Science*. Duration covers the exported session, including activities beyond reading. Valid gaze means at least one eye was marked valid. ON bundles positional restoration and highlighting.

Participant	Condition	Text	Duration (s)	Gaze samples	Rate (Hz)	Valid (%)	Changes
P1	ON	M	185.5	16,664	89.98	96.23	5
P1	OFF	N	475.1	42,683	89.90	93.94	9
P2	ON	N	252.2	22,688	90.06	98.01	7
P2	OFF	M	123.4	11,063	89.84	97.55	4
P3	ON	M	171.5	15,305	89.41	92.47	3
P3	OFF	N	385.6	34,624	89.87	95.04	11
P4	ON	N	224.0	20,136	90.00	95.20	4
P4	OFF	M	141.4	12,697	90.02	96.01	2

fraction of samples with at least one eye's gaze point marked valid. It is a device-validity measure, not accuracy of the browser's word assignment.

Three timing quantities are kept distinct. *Client diagnostic round-trip time* (RTT) is measured by browser ping/pong messages scheduled every five seconds, using the browser's millisecond clock. It includes transport and software scheduling. *Dispatch-time gaze freshness* measures elapsed backend time since the latest gaze ingestion when the backend dispatches external decision context. It does not correlate that context to a particular triggering sample. *Decision-provider RTT* measures the backend interval for correlation-matched requests and replies. Percentiles use linear interpolation over the observed samples, pooled only within a stated population.

These intervals do not together establish the time from a gaze sample to a visible intervention. In particular, analysis-window age, human approval, deliberate boundary waiting, and display presentation are not captured by the freshness measure. Its observations are dispatch records, which can outnumber distinct gaze samples. Provider RTT is conditional on a matched reply. We do not add percentiles across populations or call these separate endpoints end-to-end latency.

The integrity audit checks monotonic gaze timestamps, duplicate gaze sequence numbers and intervention identifiers, telemetry joins, and links from restoration outcomes to intervention times. For derived events, it counts repetitions of the complete inner fixation or saccade payload within each session. Such payload equality is an audit criterion, not a validated biological event-identity rule. We also recompute exploratory movement summaries after retaining only the first identical payload as a sensitivity check (Appendix A).

4.4 Advisory and geometry records

The separate advisory run lasts 91.19 s and contains 8,004 gaze samples. It exercises decision proposals and associated timing telemetry; it is not a fifth participant in the preservation comparison. Proposal histories are counted both as records and as unique proposal identifiers to avoid treating a status update as a new decision.

The browser sweep contains 66 scheduled trials per implementation version: 11 presentation changes at six page positions. The documented baseline uses a 1440×900 viewport, 18 px Merriweather text, 1.8 line height, and 680 px text width. Changes cover font size, line height, line width, letter spacing, and font family. The harness uses the production reader with gaze disabled and a viewport fallback anchor. It measures the anchor word's vertical displacement with restoration disabled and enabled. An available pair is classed as over-repositioned when the enabled displacement exceeds the disabled displacement by more than 1 px. Missing words remain missing rather than receiving zero displacement.

We analyse each saved run separately, then compare trial keys and anchor identifiers across versions. This check is essential because matching an intervention name and page position does not guarantee that both runs measured the same word or began with the same geometry. No new browser sweep or human evaluation of the revised hook was performed for this manuscript.

5 Results

5.1 Extension and researcher-control paths

The implementation supports distinct manual, analysis-provider, and decision-proposal paths. The reference providers and collaborator connector give concrete examples of the external protocol, while the 101 passing backend tests support the inspected lifecycle and persistence behaviour. In the eight recorded sessions, all 45 interventions were manually applied. The advisory run contains 16 proposal-status records for eight unique proposals: eight pending records followed by six superseded and two approved records. This distinction demonstrates why proposal history and applied intervention counts must be represented separately.

The evidence establishes exercised integration and control paths. It does not measure how quickly an unfamiliar researcher can integrate a provider, operate the console, or recover from a fault. Nor do the tests establish that provider detachment automatically substitutes a working built-in analysis strategy.

5.2 Capture and transport

Across the eight sessions, the exports contain 175,860 gaze samples. The effective sampling rate has a mean of 89.89 Hz and a sample standard deviation of 0.21 Hz, with a range of 89.41–90.06 Hz. The mean session validity is 95.56%, ranging from 92.47% to 98.01%. Exported calibration-validation summaries report average accuracy between 0.296 and 0.685 degrees and average precision between 0.083 and 0.330 degrees. These summaries concern calibration targets, not an independent validation of line or word assignment during reading.

Table 2: Observed timing endpoints in milliseconds. n counts diagnostic samples or matched exchanges, not participants. The advisory run is a separate recording. Freshness ends at context dispatch; none of these rows measures complete gaze-to-display latency.

Recording	Endpoint	n	Median	95th pct.	Maximum
Eight sessions	Participant-client RTT	387	1	3	41
Eight sessions	Researcher-client RTT	393	1	8.4	28
Advisory run	Participant-client RTT	17	1	67.8	315
Advisory run	Researcher-client RTT	18	1	18.05	24
Advisory run	Dispatch-time gaze freshness	14,015	6	11	46
Advisory run	Decision-provider RTT	8	4.5	6.3	7

For the main sessions, participant-client RTT has a median of 1 ms and a 95th percentile of 3 ms across 387 samples; the corresponding researcher-client values are 1 and 8.4 ms across 393 samples. All 780 observations are below 100 ms. In the advisory run, however, one of 17 participant-client observations is 315 ms (Table 2). The recordings therefore do not support a universal claim that every observed client exchange remained below 100 ms.

The advisory run's dispatch-time freshness has a median of 6 ms and a 95th percentile of 11 ms over 14,015 records. Eight matched decision-provider exchanges have a median RTT of 4.5 ms. These results describe the specified backend endpoints. A recent gaze sample at dispatch does not establish a recent analysis window, and neither result includes researcher response time or browser presentation.

5.3 Record integrity and derived-event repetition

All eight session/telemetry joins match, and gaze timestamps are strictly increasing with no duplicate sequence numbers. Intervention identifiers are unique within sessions. All 19 recorded preservation outcomes link to an intervention time. These checks support the internal consistency of those parts of the record.

The derived streams nevertheless contain exact repeated payloads: 280 excess fixation payloads among 9,311 records (3.0%) and 652 excess saccade payloads among 3,225 records (20.2%). Here "excess" counts repetitions after the first occurrence within a session. Thus, the exported fixation and saccade counts cannot be assumed to equal numbers of unique eye movements. The audit does not establish the original cause of these repetitions.

Retaining only the first exact saccade payload leaves the exploratory first-forward-movement medians unchanged, but changes regression-rate summaries (Appendix A). This sensitivity is consistent with the distinction between selecting the first event and counting all events within a window. It is not sufficient to validate the event stream or to justify silently replacing it with a deduplicated dataset.

5.4 Context-restoration outcomes

In the recorded ON sessions, the browser classifies three restoration outcomes as preserved and 16 as degraded. These are the hook's geometric classifications, not participant judgments of whether reading context was recovered. They show that enabling restoration did not guarantee that the implementation's own preservation criterion was satisfied.

Table 3: Exploratory browser geometry records. Displacement medians use available observations, with counts in parentheses. Over-repositioning uses complete OFF/ON pairs and a 1 px tolerance. Anchors differ in 21 of 66 trial keys across versions, precluding a matched effect estimate.

Measure	Original	Revised
Scheduled trials	66	66
Complete pairs	62	63
OFF displacement, px (n)	32.39 (62)	34.34 (63)
ON displacement, px (n)	104.38 (64)	32.39 (64)
Over-repositioned	45/62	4/63

The original browser sweep has 62 measurable pairs, with over-repositioning in 45; the revised sweep has 63 pairs, with over-repositioning in four (Table 3). These counts identify potentially useful engineering changes, but the files do not form a fully matched cross-version comparison. Of 66 corresponding trial keys, 21 have different recorded anchor token identifiers, and 26 have an OFF displacement difference greater than 1 px where both values are available. Missing pairs occur in line-width trials where the anchor word may leave the rendered page.

The available-case medians therefore describe the saved runs, not an isolated causal effect of the revision. A controlled comparison would pin the initial token and geometry across conditions, preserve missing-word outcomes, and distinguish word displacement from the hook's sentence-anchor diagnostics. The present records are useful for locating restoration problems and specifying that follow-up benchmark.

6 Discussion

6.1 Modularity needs observable experimental meaning

The architecture makes several choices independently configurable: where gaze originates, how observations are analysed, who proposes a change, and when an approved change is applied. The integration examples and recorded sessions demonstrate these boundaries in operation. For researchers, the practical value is the ability to keep one stage fixed while examining another, such as using external analysis during manual intervention sessions before enabling advisory proposals.

The evaluation also shows that an interface contract alone does not guarantee valid experimental evidence. Provider snapshots must carry a sufficiently clear notion of event identity; otherwise, a technically valid stream can repeat a movement and distort a count-based measure. Future revisions should preserve stable provider event identifiers and define whether incoming snapshots append, revise, or retransmit events. This semantics should be tested with recorded provider messages as well as serializer round trips. An inspectable record makes such defects discoverable, but does not make them harmless.

6.2 A responsive loop has several clocks and waiting points

The observed near-nominal gaze rate and low median client RTT support operation in the intended single-host setting. They do not collapse the adaptation loop into one latency number. Analysis windows, context dispatch, provider response, researcher approval, intervention commitment, and visible rendering refer to different endpoints. In an advisory experiment, some waiting is an intentional consequence of researcher control. In a boundary-scheduled experiment, waiting may also be the intervention policy itself.

Future instrumentation should correlate a triggering observation with its analysis window, proposal, committed intervention, and browser presentation measurement. It should report deliberate waiting separately from processing and transport. Direct measurement of gaze-contingent display latency, as demonstrated by Saunders and Woods [2014], provides a model for validating the visible endpoint. Without these links, a freshness metric can look favourable even when a decision rests on older features. Similarly, a sparse diagnostic ping series cannot establish the timing of every display update.

6.3 Restoration requires a declared target

The preservation example exposes a design choice that extends beyond this implementation. Maintaining the position of a word, returning to the start of a sentence, and highlighting useful context are different objectives. A restore can help one objective while moving the target used by another metric. Experiments should state which objective defines success and retain the identity and geometry of that target before and after reflow.

The present browser records motivate a benchmark with preselected tokens, pinned fonts and viewport settings, and explicit disappearance outcomes. The participant comparison would additionally need to separate positional restoration from highlighting if their effects are to be attributed independently. The current findings identify implementation behaviour and measurement requirements; they do not demonstrate that the revised hook improves reading.

6.4 Scope and limitations

The human recordings involve four convenience-sample participants, two texts, unequal intervention sequences, and a three-to-one condition order. Repeated interventions are observations nested within participants, not independent participants. No significance tests or population effect estimates are presented. The first-forward-movement proxy in the appendix is not validated against resumed comprehension or a participant report of recovery, and exact-payload repetition further limits event-rate interpretation.

Hardware evidence comes from one Tobii model and a local deployment. It does not establish performance over a network, with other trackers, or in remote or clinical use. Mouse input is a development facility rather than validation of webcam sensing. The gaze-to-word heuristic has no independent accuracy evaluation here. Automated tests cover selected backend behaviour; operator usability, independent developer integration, and systematic disconnection recovery remain unmeasured.

Historical version provenance is another limitation. Current code inspection explains the implementation under review, while the saved sessions and sweeps reflect earlier runs whose complete code, browser, font, and provider configurations are not pinned by the exports. The data hashes in the accompanying audit identify the files analysed, not their full execution environment. Replaying their events cannot remove this gap.

6.5 Privacy and ethics

Gaze trajectories, reading materials, and participant attributes in session exports can reveal sensitive information, so retaining detailed provenance also increases the importance of minimising identifying fields and controlling access. A local deployment can help researchers retain operational control, but exported records and connected providers still require an explicit data-handling protocol; this manuscript reports coded aggregates and does not redistribute raw participant records. The recruitment, consent, and institutional-review account for the existing recordings remains an author-confirmation item, and no exemption or clinical benefit is asserted.

6.6 Artifact availability and drafting disclosure

The local manuscript package includes LaTeX sources, a read-only evidence-audit script, input hashes, and generated result tables. The source review and code map identify the implementation paths behind architectural claims. This draft makes no claim that the repository or participant data has been approved for public release; a submission artifact can be prepared after version provenance and data-sharing scope are settled.

OpenAI Codex assisted with manuscript drafting, source inspection, evidence-audit code, and LaTeX preparation. The manuscript remains an internal draft for the authors to verify, particularly the participant procedure, historical configuration, interpretation of derived events, and final authorship.

7 Conclusion

Reading the Reader integrates configurable sensing and analysis with researcher-controlled interventions and a record of the resulting experiment. Its contribution is the connection between these extension boundaries and an inspectable reading workflow. Existing sessions demonstrate near-nominal 90 Hz capture, manually applied changes, and separately exercised advisory proposals in a single-host deployment. The audit also exposes repeated derived-event payloads, incomplete timing coverage, and restoration outcomes that require more precise targets and matching. Together, the implementation and evaluation provide a concrete foundation for further gaze-contingent reading experiments while defining the validation needed before claims about reading benefit or complete intervention latency.

References

Sachin Baral and Satish Gurung. 2026. *Reading the Reader: Adaptive Reading Systems: A Modular Software Architecture.* Master's thesis. Technical University of Denmark.

Chaudhary Muhammad Aqdus Ilyas, Sifat-E Noor, Ashkan Tashk, Bart Cooreman, Sofie Beier, and Per Bækgaard. 2025. Reading the Readers Mind through Eye Tracking: Can AI Generated Texts Match Human Authors?. In *Proceedings of the 2025 Symposium on Eye Tracking Research and Applications (ETRA '25)*. Association for Computing Machinery, New York, NY, USA, Article 113, 7 pages. doi:10.1145/3715669.3726846

Helena Eschricht Jensen, Chaudhary Muhammad Aqdus Ilyas, Ashkan Tashk, Bart Cooreman, Sofie Beier, and Per Bækgaard. 2025. Context Preservation Through Eye Tracking: Adaptive Reading Application Design for an Optimal Reading Experience. In *Proceedings of the 2025 Symposium on Eye Tracking Research and Applications (ETRA '25)*. Association for Computing Machinery, New York, NY, USA, Article 78, 8 pages. doi:10.1145/3715669.3725897

Franziska Kaltenberger, Wei-Ling Chen, Enkeleda Thaqi, and Enkelejda Kasneci. 2026. Sure About That Line? Approaching Confidence-Based, Real-Time Line Assignment in Reading Gaze Data. *Proceedings of the ACM on Computer Graphics and Interactive Techniques* 9, 2 (May 2026). doi:10.1145/3803540

Sebastiaan Mathôt, Daniel Schreij, and Jan Theeuwes. 2012. OpenSesame: An Open-Source, Graphical Experiment Builder for the Social Sciences. *Behavior Research Methods* 44, 2 (2012), 314–324. doi:10.3758/s13428-011-0168-7

Elad Medan and Barak Pelman. 2024. New Line: A New Gaze-Contingent Digital Reading Interface. Paper hosted by the MEITAL 2024 conference. https://meitalconf.iucc.ac.il/wp-content/uploads/2024/08/MeitalConference_2024_paper_14.pdf

Jonathan Peirce, Jeremy R. Gray, Sol Simpson, Michael MacAskill, Richard Höchenberger, Hiroyuki Sogo, Erik Kastman, and Jonas Kristoffer Lindeløv. 2019. PsychoPy2: Experiments in Behavior Made Easy. *Behavior Research Methods* 51, 1 (2019), 195–203. doi:10.3758/s13428-018-01193-y

PsychoPy contributors. 2026. ioHub Common Eye Tracker Interface. PsychoPy documentation. https://devdocs.psychopy.org/api/iohub/device/eyetracker.html

Koen Rummens and Sofie Beier. 2026. Gaze-based word highlighting boosts reading performance: an eye tracking study in second graders. *Ergonomics* 69, 7 (2026), 1253–1265. doi:10.1080/00140139.2025.2514600 First published online 12 June 2025.

Daniel R. Saunders and Russell L. Woods. 2014. Direct Measurement of the System Latency of Gaze-Contingent Displays. *Behavior Research Methods* 46, 2 (2014), 439–447. doi:10.3758/s13428-013-0375-5

A Exploratory Post-intervention Movement Analysis

This appendix retains the descriptive movement analysis as an example of what can be computed from the recorded intervention timeline. It is not an efficacy evaluation. For an intervention at epoch time t_i, the first-forward-movement delay is the earliest recorded saccade start after t_i whose direction is forward or line-change-forward, minus t_i. The search ends at the next intervention or 30 s, whichever is earlier. The estimator falls back to the event occurrence time if a start time is absent. If no qualifying event is found, the observation remains missing. This operational definition does not establish that comprehension or stable reading resumed at that movement.

Table 4: Participant-level first-forward-movement medians (ms), with observed / applied counts in parentheses. Texts and intervention sequences differ across conditions. P3 has one missing OFF observation.

Participant	ON: ms (n/N)	OFF: ms (n/N)
P1	580 (5/5)	613 (9/9)
P2	735 (7/7)	1178.5 (4/4)
P3	482 (3/3)	350 (10/11)
P4	259 (4/4)	731 (2/2)

The pooled event medians are 482 ms for ON (19 of 19 observations) and 650 ms for OFF (25 of 26). Three participants have a lower ON median; one has a lower OFF median (Table 4). Pooling is descriptive: the 44 observed delays are not 44 independent participants, and the conditions bundle restoration with highlighting. These medians should not be interpreted as a validated reduction in reading-recovery time.

For regression summaries, the estimator computes the fraction of recorded saccades marked as regressions in five-second windows before and after each intervention, truncated at neighbouring interventions. Condition summaries average the fractions of nonempty windows rather than pooling all saccades. On the original payload stream, the ON summaries are 29.30% before and 22.76% after; OFF summaries are 31.96% and 33.01%. Keeping only the first identical inner saccade payload changes these to 30.88% and 23.02% for ON, and 32.69% and 33.02% for OFF. The first-forward-movement medians remain unchanged under that sensitivity check. Exact-payload removal is not a validated correction of event identity, so neither representation is promoted to a canonical behavioural result.

The audit also provides three- and ten-second window summaries. Their descriptive direction is consistent with the five-second condition-level summaries, but effect magnitudes vary. Resolving event identity, recording a reproducible protocol, and validating the recovery measure would be prerequisites for an intervention-effect study.